

ENGINEERING SPECIFICATION

NomadSync

The Ultimate Multi-Paradigm React Native Learning Blueprint

Target Audience: Mobile Software Engineers

Core Tech Stack: React Native, Expo (Prebuild), SQLite, Swift, Kotlin

Architecture: Offline-First, Event-Driven, Native Bridged

Document Ver: v1.0.0 (Production-Ready Layout)

Project Overview & Learning Scope

The **NomadSync** application is an architectural exercise designed to cover the entire spectrum of mobile platform development. Standard clone apps typically limit learners to isolated concepts: an Instagram clone explores high-volume media feeds and layout streaming; an Uber clone exercises foreground geospatial positioning and fast socket connections.

NomadSync intentionally merges these paradigms into a production-grade **Field Tracking & Media Synchronization platform**. By implementing this blueprint, developers will transition from executing standard UI component layouts to managing low-level mobile runtime complexities.

Core Native Competencies Gained

By building NomadSync, you will cross-compile features directly targeting background task priority loops, persistent hardware keychain buffers, image down-sampling logic on the device background thread, and local differential sync protocols via SQLite triggers.

Global Core Architecture Specs

To ensure complete mastery, the underlying application layout must adhere to an **Offline-First architecture**. Every UI button action must commit instantly to local state storage before network attempts are initialized.

- **Local Persistence Engine:** SQLite integrated via a persistent reactive abstraction layer. No raw runtime keys stored dynamically via high-latency string storage.
- **Network Synchronization Strategy:** Atomic transaction queuing. Network operations leverage an explicit queue structure tracking state flags: [pending, syncing, failed].
- **Media Optimization Pipeline:** Multi-threaded image canvas compression. Photos captured at high-resolution form an background operational thread utilizing hardware-accelerated downscaling parameters.

Page 1: Auth & Biometric Gateway

The initial launch screen manages cryptographic device-level enrollment. Rather than handling state profiles inside vanilla runtime contexts, auth tokens are structurally isolated inside native hardware containers.

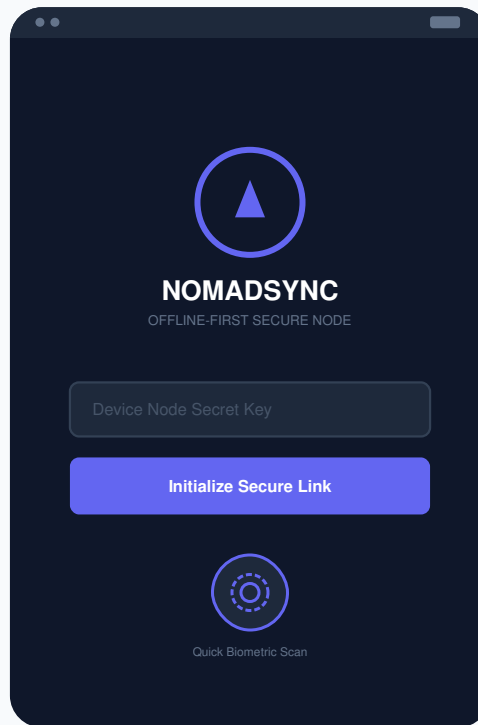


Figure 1.1: Authentication & Node Access Controller UI Layout (Dark Theme Matrix)

Functional Requirements & Behavioral Matrix

Component ID	Trigger Action	Subsystem Functional Execution Pipeline
Input-A1 (Secret Field)	Focus / Input character streams	Captures raw text arrays. Prevents cache generation by explicitly declaring secure text entry parameters to the host system OS layer.
Btn-A1 (Initialize Link)	OnPress Event	Dispatches text to a cryptographic hashing loop. On confirmation, writes the generated private token payload into the physical platform Keychain (iOS) or Keystore (Android) via an asymmetric native bridge block. Transitions navigation layout index root smoothly.
Btn-A2 (Biometric Key)	OnPress Event	Suspends the JavaScript application loop and invokes the hardware OS local auth daemon. Triggers native hardware prompt interfaces. On valid biometric parsing, decrypts the token storage map dynamically and passes app instantiation permissions.

Page 2: Core Telemetry & Geospatial Dashboard

The interface center governing tracking vectors. This page relies heavily on complex multi-layer Mapbox/Google map views, drawing rendering lines dynamically while simultaneously listening to a continuous background location service thread.

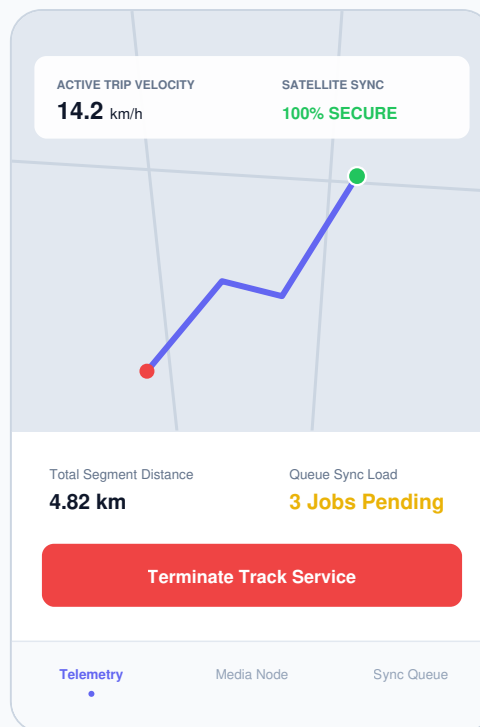


Figure 1.2: Real-time Geospatial Telemetry Core Dashboard Interface (Light Mode Rendering)

Functional Requirements & Behavioral Matrix

Component ID	Trigger Action	Subsystem Functional Execution Pipeline
Map-Canvas	OnRegionChange Event	Renders map tiles matching vector bounds. Bound paths listen directly to the local reactive schema database collection to update coordinates seamlessly.
Btn-B1 (Service Switch)	OnPress Event	Evaluates conditional state switches. When activated to terminate, it unlinks the native platform CLLocationManager (iOS) or FusedLocationProviderClient (Android) hooks. Kills the background thread lifecycle instance smoothly and flushes pending coordinates to local storage.
Tab-Navigation Links	OnPress Navigation Element	Executes standard state routing instructions without purging background operational loops. Switches target layout blocks instantly.

Page 3: Smart Media Node Capture Feed

This interface block replaces typical social image views with an intelligent media-sourcing component. It bridges directly to custom device camera layers to capture raw visual data and automatically binds contextual geospatial metadata fields to the file header.

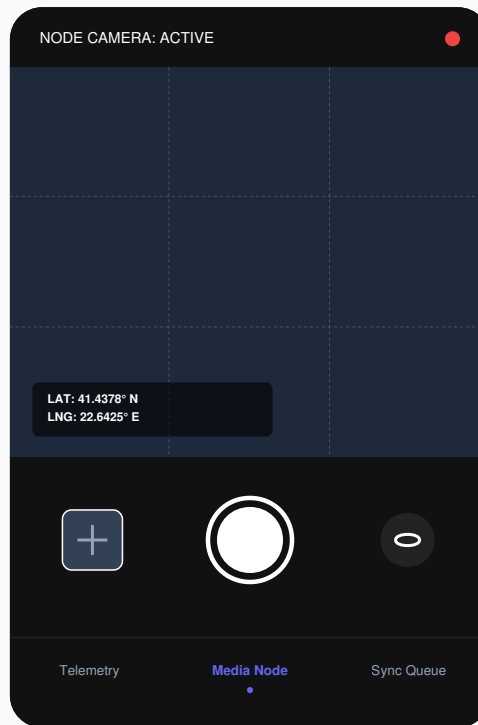


Figure 1.3: Precision Camera Node Interface with Real-time GPS Tagging Matrix

Functional Requirements & Behavioral Matrix

Component ID	Trigger Action	Subsystem Functional Execution Pipeline
Btn-C1 (Gallery Preview)	OnPress Event	Invokes local device photo repository routing layers, opening up existing localized file instances instantly without disturbing active map background sockets.
Btn-C2 (Shutter Core)	OnPress Event	Executes asymmetric capture sequences: 1. Resolves high-res raw image data string streams. 2. Pulls instantaneous location coordinate variables via the geospatial hardware tracker. 3. Intercepts local image files, writing coordinates directly into structural metadata maps. 4. Stores paths securely to the device database while queuing synchronization flags.
Btn-C3 (Toggle Lens)	OnPress Event	Reconfigures hardware camera state parameters. Swaps logical input streams smoothly between device front arrays and ultra-wide back lenses.

Page 4: Transaction & Synchronization Queue Monitor

The core engine inspection interface. Rather than treating remote API transmission as background magic, this view displays the tracking queue blocks, offering deep visibility into transactional synchronization pipelines.

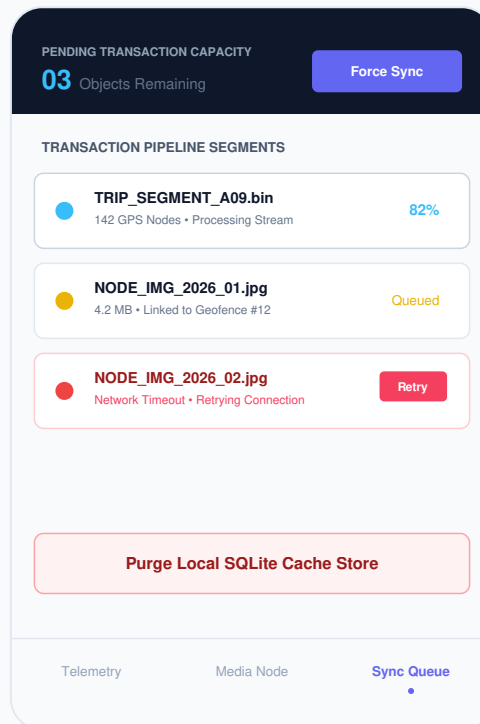


Figure 1.4: Real-time Data Synchronization Queue Engine Inspector View

Functional Requirements & Behavioral Matrix

Component ID	Trigger Action	Subsystem Functional Execution Pipeline
Btn-D1 (Force Sync)	OnPress Event	Overrides current network interval back-off timings. Immediately fetches outstanding local mutations and initializes parallel multipart binary streaming hooks to the target remote backend API.
Btn-D2 (Row Retry)	OnPress Event	Isolates the target record index item inside the local transaction table. Strips existing error tracking states and re-inserts the entity back into the active synchronization loop.
Btn-D3 (Purge Store)	OnLongPress Event	Executes system cache clearing safety checks: 1. Displays system security native prompt confirms. 2. Runs DROP TABLE / DELETE FROM sweeps over local storage engines. 3. Resets live application operational variables down to clean zero-states.